Question 6.17(2) on Unconditional FDDs

Audit, literature status, and an exact tail-localization criterion

Research report prepared from the supplied paper and partial solution

22 June 2026

Abstract

The question is whether every non-Hilbertian Banach space with an unconditional finite-dimensional decomposition (UFDD) contains a non-Hilbertian subspace with an unconditional basis. I did not locate a published proof or counterexample. The proposed sufficient condition is correct after repairing one uniformity argument, but its description in terms of Pisier's property (H) is inaccurate. More significantly, I prove a converse: for a fixed UFDD, the proposed finite-dimensional tail-witness condition is equivalent to the desired conclusion. Thus the partial result is an exact localization of the open problem rather than a strictly weaker theorem. I also reduce the hard case to a non-Hilbertian Hilbertian sum $(\bigoplus E_n)_2$ of finite-dimensional spaces and prove several positive cases. In particular, any counterexample must have type p for every $p < 2$ and cotype q for every $q > 2$.

1 The source question and the verdict

The source is W. Cuellar Carrera, N. de Rancourt and V. Ferenczi, *Local Banach-space dichotomies and ergodic spaces* [5]. Question 6.17(2) asks:

> Does every non-Hilbertian space with an unconditional FDD have a non-Hilbertian subspace with an unconditional basis?

The bibliographic attribution in the supplied packet is incorrect: the paper is not by Ferenczi–Lopez-Abad–Mbombo–Todorcevic. It appeared in the *Journal of the European Mathematical Society* in 2023; the preprint is arXiv:2005.06458.

The outcome of the investigation is as follows.

1. The theorem in the packet is correct, but its proof of a uniform ℓ_2 estimate has a logical gap. A corrected tail version is proved below and suffices for the concatenation argument.

2. The phrase "uniform tail property-(H) failure" should be avoided. The condition concerns Banach–Mazur distance of finite-dimensional spaces, while Pisier's property (H) controls equal-coefficient sums of unconditional sequences. They are not equivalent. Non-Hilbertian weak Hilbert spaces with unconditional bases show that the proposed tail condition can coexist with property (H) [11, 10, 2].

3. Exact-wording searches, citation trails, and the classical and recent literature on UFDDs, weak Hilbert spaces, twisted Hilbert spaces and d_2-HI spaces did not reveal a full answer or counterexample through the search date.

4. Relative to a fixed UFDD, the packet's hypothesis is not merely sufficient: it is equivalent to the existence of the desired subspace. This exact tail-localization is the main additional result below.

Throughout, spaces are real Banach spaces. For a finite-dimensional space F, write

$$d_2(F) = d_{\mathrm{BM}}(F, \ell_2^{\dim F}).$$

A normalized basic sequence is K-unconditional when arbitrary sign changes have norm at most K times the original norm.

2 A finite-dimensional estimate

Lemma 2.1 (Unconditional Euclidean comparison)**.** *Let G be m-dimensional and let $(x_i)_{i=1}^m$ be a normalized K-unconditional basis of G. For every $D > d_2(G)$, the sequence (x_i) is KD-equivalent to the unit vector basis of ℓ_2^m:*

$$\frac{1}{KD}\Big(\sum_{i=1}^m |a_i|^2\Big)^{1/2} \le \Big\|\sum_{i=1}^m a_i x_i\Big\| \le KD\Big(\sum_{i=1}^m |a_i|^2\Big)^{1/2}.$$

Proof. Choose $T : G \to \ell_2^m$ with $\|T\|\,\|T^{-1}\| < D$. Put $x = \sum_i a_i x_i$ and let (ε_i) be independent signs. Unconditionality gives

$$K^{-1}\|x\| \le \Big(\mathbb{E}\big\|\sum_i \varepsilon_i a_i x_i\big\|^2\Big)^{1/2} \le K\|x\|.$$

Orthogonality in ℓ_2^m gives

$$\frac{1}{\|T\|\|T^{-1}\|}\|a\|_2 \le \Big(\mathbb{E}\big\|\sum_i \varepsilon_i a_i x_i\big\|^2\Big)^{1/2} \le \|T\|\|T^{-1}\|\|a\|_2.$$

Combining the inequalities proves the assertion. □

Remark 2.2. Large $d_2(G)$ need not force the equal-sum vector of a particular unconditional basis of G to have abnormal size. Hence the packet's condition should be renamed rather than identified with failure of property (H).

3 The corrected sufficient theorem

Let (E_n) be a UFDD of X, and write $P_{[r,s]}$ for its interval projections.

Definition 3.1 (Tail-witness property)**.** The UFDD (E_n) has the *tail-witness property* if there is $\lambda \ge 1$ such that, for every $N \in \mathbb{N}$ and $M \ge 1$, there are $N \le r \le s$ and

$$G \subseteq E_r \oplus \cdots \oplus E_s$$

with a normalized λ-unconditional basis and $d_2(G) > M$.

The next lemma repairs the gap in the packet.

Lemma 3.2 (Uniformity on a tail)**.** *Assume every normalized block sequence of (E_n) spans a Hilbertian space. Then there are $N_0 \in \mathbb{N}$ and $C \ge 1$ such that every normalized block sequence supported after N_0 is C-equivalent to the unit vector basis of ℓ_2.*

Proof. Suppose not. Recursively choose finite normalized block sequences

$$(u_i^{(j)})_{i=1}^{m_j}, \qquad j = 1, 2, \ldots,$$

with successive, mutually disjoint supports, such that the jth sequence is not j-equivalent to the unit vector basis of $\ell_2^{m_j}$. Their concatenation is a normalized block sequence. It is unconditional and, by hypothesis, spans a Hilbertian space. A normalized unconditional basic sequence in a Hilbert space is equivalent to the ℓ_2 basis with one fixed constant. The same constant works on every finite consecutive piece, which contradicts the choice of the pieces. □

The packet asserted a global constant and argued that failure of global uniformity supplies bad sequences in every tail. That implication was not proved. Lemma 3.2 is the direct gliding-hump conclusion and is all that the proof needs. A global estimate can subsequently be recovered by separating the uniformly bounded finite head from the controlled tail, but that extra step was absent from the packet.

Theorem 3.3 (Corrected packet theorem)**.** *Let X have a UFDD (E_n). If (E_n) has the tail-witness property, then X contains a non-Hilbertian subspace with an unconditional basis.*

Proof. If some normalized block sequence spans a non-Hilbertian space, it already is an unconditional basic sequence. Otherwise take N_0, C from Lemma 3.2.

Choose successive interval-supported spaces G_k, all after N_0, such that $d_2(G_k) \to \infty$ and each G_k has a normalized λ-unconditional basis $(x_i)_{i_k \le i < i_{k+1}}$. For finitely supported scalars (a_i) and signs (ε_i), put

$$u_k = \sum_{i_k \le i < i_{k+1}} a_i x_i, \qquad v_k = \sum_{i_k \le i < i_{k+1}} \varepsilon_i a_i x_i,$$

and let $b_k = \|u_k\|$, $c_k = \|v_k\|$. When nonzero, write $y_k = u_k/b_k$ and $z_k = v_k/c_k$. Internal unconditionality gives $c_k \le \lambda b_k$, while (y_k) and (z_k) are normalized FDD block sequences after N_0. Hence

$$\begin{aligned}\Big\|\sum_i \varepsilon_i a_i x_i\Big\| &= \Big\|\sum_k c_k z_k\Big\| \\ &\le C\Big(\sum_k c_k^2\Big)^{1/2} \le \lambda C\Big(\sum_k b_k^2\Big)^{1/2} \\ &\le \lambda C^2\Big\|\sum_k b_k y_k\Big\| = \lambda C^2\Big\|\sum_i a_i x_i\Big\|.\end{aligned}$$

Thus the concatenation is unconditional. Its closed span contains every G_k and therefore cannot be Hilbertian, since $d_2(G_k) \to \infty$. □

4 A converse: exact tail localization

The converse uses a finite-rank avoidance lemma.

Lemma 4.1 (Annihilating a finite-rank operator)**.** *Let Y be a non-Hilbertian space with a normalized unconditional basis. There is $L \ge 1$, depending only on that basis, such that for every finite-rank $T : Y \to F$, every $\eta > 0$, and every $M \ge 1$, there is a finite-dimensional $G \subseteq Y$ satisfying:*

(i) G has a normalized L-unconditional basis;

(ii) $d_2(G) > M$;

(iii) $\|T|_G\| < \eta$.

Proof. Let (y_j) be a normalized K-unconditional basis of Y. We use the standard fact that a nonreflexive space with an unconditional basis contains a normalized block sequence equivalent to the canonical basis of either c_0 or ℓ_1 [1]. Every normalized block sequence of (y_j) is K-unconditional, so $L = K$ works in all cases.

Reflexive case. The basis is shrinking. For the tail projection $Q_m : Y \to [y_j : j \geq m]$, one has $\|TQ_m\| \to 0$ because T has finite rank. Every basis tail is non-Hilbertian. Consequently

$$d_2([y_j : m \leq j \leq r]), \qquad r \geq m,$$

is unbounded: otherwise Lemma 2.1 would make the entire tail basis uniformly equivalent to ℓ_2. Choose m with $\|TQ_m\| < \eta$, then choose a coordinate interval G in that tail with $d_2(G) > M$.

c_0 case. Fix a normalized block sequence (w_j) which is D-equivalent to the c_0 basis. It is weakly null, so $\|Tw_j\| \to 0$. Pass to a subsequence with

$$\sum_j \|Tw_j\| < \eta/D.$$

For $x = \sum_{j=1}^m a_j w_j$ we have $\max_j |a_j| \leq D\|x\|$, and hence $\|Tx\| < \eta\|x\|$. For $G_m = [w_1, \dots, w_m]$,

$$d_2(G_m) \geq D^{-2} d_{\mathrm{BM}}(\ell_\infty^m, \ell_2^m) = D^{-2}\sqrt{m}.$$

Take m large.

ℓ_1 case. Fix a normalized block sequence (v_j) which is D-equivalent to the ℓ_1 basis. Since F is finite-dimensional, (Tv_j) has a Cauchy subsequence. Choose successive disjoint pairs from it so rapidly that, after normalization,

$$w_i = \frac{v_{p_{2i}} - v_{p_{2i-1}}}{\|v_{p_{2i}} - v_{p_{2i-1}}\|}, \qquad \sum_i \|Tw_i\| < \eta/D_1,$$

where (w_i) is D_1-equivalent to the ℓ_1 basis for a fixed D_1. Then $\|T|_{[w_1,\dots,w_m]}\| < \eta$ and

$$d_2([w_1, \dots, w_m]) \geq D_1^{-2} d_{\mathrm{BM}}(\ell_1^m, \ell_2^m) = D_1^{-2}\sqrt{m}.$$

Again take m large. □

Theorem 4.2 (Exact tail-localization criterion)**.** *Let X have a UFDD (E_n). The following are equivalent.*

(a) X contains a non-Hilbertian subspace with an unconditional basis.

(b) (E_n) has the tail-witness property.

Proof. The implication (b)⇒(a) is Theorem 3.3.

For the converse, let $Y \subseteq X$ be non-Hilbertian with an unconditional basis. Fix N and apply Lemma 4.1 to $T = P_{[1,N-1]}|_Y$. Let $\delta = 1/8$. For a number R to be chosen, obtain $G \subseteq Y$ with a normalized L-unconditional basis, $d_2(G) > R$, and

$$\|P_{[1,N-1]}|_G\| < \delta.$$

Because G is finite-dimensional, choose $s \geq N$ so that $\|P_{(s,\infty)}|_G\| < \delta$. Put $S = P_{[N,s]}|_G$. Then $\|S - \mathrm{Id}_G\| < 2\delta < 1$, so S is an isomorphism onto its image and

$$\|S\|\,\|S^{-1}\| \leq \frac{1+2\delta}{1-2\delta} =: D_0.$$

The image SG lies in $E_N \oplus \cdots \oplus E_s$. After normalizing the image basis, it is LD_0-unconditional, and

$$d_2(SG) \geq \frac{d_2(G)}{D_0}.$$

Choosing $R > D_0 M$ proves the tail-witness property with the fixed constant $\lambda = LD_0$. $\square$

Remark 4.3. The packet's hypothesis is therefore exactly the desired conclusion localized into arbitrarily remote finite intervals of the fixed UFDD. This validates the mechanism, but also shows that it does not by itself reduce the open problem to a genuinely weaker statement.

5 The hard case: a Hilbertian sum of bad blocks

Theorem 5.1 (Block/Hilbertian-sum reduction)**.** *Let X be a non-Hilbertian space with a UFDD (E_n). Then one of the following holds.*

(i) Some normalized block sequence of (E_n) spans a non-Hilbertian space; hence the question has a positive answer for X.

(ii) There are N_0 and C such that, for all finitely supported $x_n \in E_n$ with $n \geq N_0$,

$$C^{-1}\Big(\sum_{n\geq N_0} \|x_n\|^2\Big)^{1/2} \leq \Big\|\sum_{n\geq N_0} x_n\Big\| \leq C\Big(\sum_{n\geq N_0} \|x_n\|^2\Big)^{1/2}.$$

Consequently the tail is isomorphic to $(\bigoplus_{n\geq N_0} E_n)_2$. Moreover, for every $N \geq N_0$,

$$\sup_{n\geq N} d_2(E_n) = \infty.$$

Proof. If (i) fails, Lemma 3.2 gives N_0, C. Apply its C-equivalence estimate to the normalized one-block vectors $x_n/\|x_n\|$ to obtain the displayed inequality.

If $\sup_{n\geq N} d_2(E_n) < \infty$ on some tail, choose uniformly controlled isomorphisms of those E_n onto Euclidean spaces and take their Hilbertian direct sum. The displayed estimate would make that tail Hilbertian. Adding finitely many earlier summands would then make X Hilbertian, a contradiction. $\square$

In case (ii), the tail is reflexive and ℓ_2-saturated. The unresolved issue is internal to the finite-dimensional blocks: their Euclidean distortion is unbounded, but uniformly unconditional non-Euclidean subspaces might become harder to find farther out.

For $\lambda \geq 1$ and $N \in \mathbb{N}$, define

$$\Phi_\lambda(N) = \sup\left\{ d_2(G) : \begin{array}{l} G \subseteq E_r \oplus \cdots \oplus E_s, \quad N \leq r \leq s, \\ G \text{ has a normalized } \lambda\text{-unconditional basis} \end{array} \right\}.$$

By Theorem 4.2, the answer for X is positive exactly when $\Phi_\lambda(N) = \infty$ for some fixed λ and every N.

6 Concrete positive consequences

Corollary 6.1 (Bounded-dimensional UFDD)**.** *If X is non-Hilbertian and has a UFDD (E_n) with $\sup_n \dim E_n < \infty$, then X contains a non-Hilbertian block subspace with an unconditional basis.*

Proof. John's theorem gives $d_2(E_n) \leq \sqrt{\dim E_n}$ uniformly [1]. Alternative (ii) of Theorem 5.1 is impossible. $\square$

Thus classical examples with a two-dimensional UFDD but no unconditional basis for the whole space, including the Kalton–Peck and Komorowski-type examples, are not counterexamples to the subspace question [4, 6, 8].

Corollary 6.2 (Uniformly unconditional blocks)**.** *Suppose X is non-Hilbertian with a UFDD (E_n), and every E_n has a normalized K-unconditional basis for one fixed K. Then X contains a non-Hilbertian subspace with an unconditional basis.*

Proof. In alternative (i) of Theorem 5.1 we are done. In alternative (ii), the numbers $d_2(E_n)$ are unbounded on every tail, so the individual blocks furnish the tail-witness property with $\lambda = K$. □

Corollary 6.3 (Uniformly Euclidean blocks)**.** *Suppose X has a UFDD (E_n) satisfying*

$$\sup_n d_2(E_n) < \infty.$$

If X is non-Hilbertian, then it contains a non-Hilbertian block subspace with an unconditional basis.

Proof. Alternative (ii) of Theorem 5.1 is impossible. □

Corollary 6.4 (Type–cotype obstruction)**.** *For a Banach space X, put*

$$p_X = \sup\{p \in [1,2] : X \text{ has type } p\}, \qquad q_X = \inf\{q \in [2,\infty) : X \text{ has cotype } q\},$$

with $q_X = \infty$ if X has no finite cotype. Let X be non-Hilbertian and have a UFDD. If $p_X < 2$ or $q_X > 2$, then X contains a non-Hilbertian subspace with an unconditional basis. Consequently, every counterexample would satisfy

$$p_X = q_X = 2;$$

equivalently, it would have type p for every $p < 2$ and cotype q for every $q > 2$, without necessarily having type 2 or cotype 2.

Proof. Finite-dimensional perturbations do not change type or cotype exponents, so every UFDD tail has the same indices. The Maurey–Pisier theorem says that ℓ_{p_X} is finitely representable in every tail and, if $q_X > 2$, ℓ_{q_X} is finitely representable there; when $q_X = \infty$, use ℓ_∞ [9].

Assume $p_X < 2$. For every tail and m, choose a subspace H which is 2-isomorphic to $\ell_{p_X}^m$. Since H is finite-dimensional, a long enough interval projection is a 2-isomorphism on H. Its image G is supported in one finite tail interval, has a normalized unconditional basis with a universal constant, and

$$d_2(G) \geq \frac{1}{4} d_{\mathrm{BM}}(\ell_{p_X}^m, \ell_2^m) = \frac{1}{4} m^{1/p_X - 1/2} \longrightarrow \infty.$$

Thus the tail-witness property holds. The case $2 < q_X < \infty$ is identical, using $d_{\mathrm{BM}}(\ell_{q_X}^m, \ell_2^m) = m^{1/2 - 1/q_X}$, and for $q_X = \infty$ use $d_{\mathrm{BM}}(\ell_\infty^m, \ell_2^m) = \sqrt{m}$. □

Casazza–Kalton proved that a bounded-dimensional UFDD together with Gordon–Lewis local unconditional structure yields an unconditional basis for the whole space [4]. Corollary 6.1 assumes no local unconditional structure, but concludes only for a suitable non-Hilbertian subspace.

7 Literature status

The source paper proves a positive answer for d_2-minimal spaces (Theorem 6.15 of [5]). Its proof obtains uniformly unconditional, increasingly non-Euclidean finite-dimensional witnesses from d_2-minimality and failure of property (H), then uses essentially the concatenation mechanism of the supplied packet.

Casazza–Kalton [4] is the closest classical paper on assembling bases inside UFDD summands. It does not settle the present subspace problem for unbounded block dimensions. Kalton proved that a twisted Hilbert space with an unconditional basis is Hilbertian, and his argument also controls complemented unconditional basic sequences [7]; this does not rule out arbitrary uncomplemented subspaces of a twisted Hilbert space.

The 2025 paper of Argyros–Manoussakis–Motakis constructs classical models related to the announced ℓ_2-saturated d_2-HI problem, but explicitly leaves the exotic construction for future work [3]. The recent work of de Rancourt–Kurka proves ergodicity for non-Hilbertian Orlicz sequence spaces and several twisted Hilbert spaces, not Question 6.17(2) [12].

Exact searches for the wording of Question 6.17(2), searches combining "non-Hilbertian", "unconditional FDD", "unconditional basis", "property (H)", and "twisted Hilbert", and inspection of the citation trail of the JEMS paper did not reveal a full solution. The responsible conclusion is therefore that no full answer was located and the question appears to remain open. This is a literature-search conclusion, not a claim about unpublished work.

8 What a counterexample must accomplish

By Theorems 4.2 and 5.1, after deleting finitely many coordinates a counterexample may be assumed to be

$$X = \Big(\bigoplus_{n=1}^{\infty} E_n\Big)_2,$$

where $\sup_n d_2(E_n) = \infty$. By Corollary 6.4, it must also satisfy $p_X = q_X = 2$. Nevertheless, for every fixed λ, some tail must impose a uniform upper bound on $d_2(G)$ for every finite-dimensional interval-supported G with a normalized λ-unconditional basis.

Thus choosing individual blocks with large unconditional-basis constants is not enough: one must prevent uniformly unconditional, non-Euclidean subspaces from emerging across several successive blocks of the Hilbertian sum. Random finite-dimensional spaces and spaces with poor local unconditional structure are natural candidates, but stability of the required obstruction under finite Hilbertian sums is the missing step.

Conversely, a positive solution is equivalent to the following concrete finite-dimensional principle.

Question 8.1 (Residual finite-dimensional principle)**.** *Let (E_n) be finite-dimensional spaces such that $(\bigoplus_n E_n)_2$ is non-Hilbertian. Must there exist a constant λ and successive finite intervals I_k containing subspaces*

$$G_k \subseteq \Big(\bigoplus_{n\in I_k} E_n\Big)_2$$

with normalized λ-unconditional bases and $d_2(G_k) \to \infty$?

Question 8.1 is equivalent to Question 6.17(2) through the block reduction and the exact tail-localization theorem. It is the sharp endpoint reached here.

9 Conclusion

The automatic solver found a valid concatenation mechanism. Its theorem is sound after two revisions: replace the flawed global-uniformity argument by Lemma 3.2, and replace the property-(H) terminology by the tail-witness terminology. The principal extension is Theorem 4.2, which proves that the packet's condition is necessary as well as sufficient. The hard case is reduced to a finite-dimensional extraction problem in non-Hilbertian Hilbertian sums, and several substantial classes are ruled out as counterexamples. I did not obtain a complete proof or a counterexample.

References

[1] F. Albiac and N. J. Kalton, *Topics in Banach Space Theory*, 2nd ed., Graduate Texts in Mathematics 233, Springer, 2016.

[2] S. A. Argyros, K. Beanland and T. Raikoftsalis, A weak Hilbert space with few symmetries, *C. R. Math. Acad. Sci. Paris* **348** (2010), 1293–1296.

[3] S. A. Argyros, A. Manoussakis and P. Motakis, Variants of the James tree space, *Israel J. Math.* **265** (2025), 1–77.

[4] P. G. Casazza and N. J. Kalton, Unconditional bases and unconditional finite-dimensional decompositions in Banach spaces, *Israel J. Math.* **95** (1996), 349–373.

[5] W. Cuellar Carrera, N. de Rancourt and V. Ferenczi, Local Banach-space dichotomies and ergodic spaces, *J. Eur. Math. Soc.* **25** (2023), 3537–3598.

[6] N. J. Kalton and N. T. Peck, Twisted sums of sequence spaces and the three-space problem, *Trans. Amer. Math. Soc.* **255** (1979), 1–30.

[7] N. J. Kalton, Twisted Hilbert spaces and unconditional structure, *J. Inst. Math. Jussieu* **2** (2003), 401–408.

[8] R. Komorowski, On constructing Banach spaces with no unconditional basis, *Proc. Amer. Math. Soc.* **120** (1994), 101–107.

[9] B. Maurey and G. Pisier, Séries de variables aléatoires vectorielles indépendantes et propriétés géométriques des espaces de Banach, *Studia Math.* **58** (1976), 45–90.

[10] N. J. Nielsen and N. Tomczak-Jaegermann, Banach lattices with property (H) and weak Hilbert spaces, *Illinois J. Math.* **36** (1992), 345–371.

[11] G. Pisier, Weak Hilbert spaces, *Proc. London Math. Soc.* (3) **56** (1988), 547–579.

[12] N. de Rancourt and O. Kurka, The ergodicity of Orlicz sequence spaces, *J. Funct. Anal.* **290** (2026), no. 5, 111270.